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THIRD SEMESTER **B.Tech.**

END SEMESTER EXAMINATION ***(Nov -2016)***

BT/EC/EE/EL/EN/CO/IT/SE -201- ENGINEERING ECONOMICS

Time: 3 Hour ***Max. Marks: 50***

Note: Answer any five questions. Assume suitable missing data, if any.

1.a Discuss Production Function and its relevance for Engineers in the globalised economic environment. 5

1.b A person is planning for his retired life. He has 10 more years of service. He would like to deposit Rs. 8500 at the end of first year and thereafter, he wishes to deposit the amount with an annual decrease of Rs. 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series. 5

2.a Discuss Business Cycle. Discuss role of engineer in bringing out the economy out of recession. 5

2.b A concrete aggregate mix must contain at least 31% sand by volume for proper batching. One source of material, which has 25% sand and 75% coarse aggregate, sells for Rs 3 per cubic meter (m^3). Another source, which has 40% sand and 60% coarse aggregate, sells for Rs. 4.40 per cubic meter (m^3). Determine the least cost per cubic meter of blended aggregates. 2

2.c If the demand function is given by x=20/(p+1), find the Price Elasticity of Demand at the point where p=3. 3

3.a Discuss five salient features of the India economy. 5

3. b A computer producing company can either purchase the screen or can manufacture within the company. The Detail is given below 5

	Purchase	Manufacture
Price per screen	4000	
Tax rate	@ 10% per unit	
Cost of machine		20 lakh
Cost of labour		2000/unit
Cost of raw material		1000/unit
Overhead Cost		400/ unit

If the annual demand is 2200, the company should make or buy.

4.a Recently Rs 1000 and Rs.500 notes have been banned in India and not a legal tender any more. Discuss its implication on the Indian economy? 5

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4.b A transport company has been looking for a new tyre for its truck and has located the following alternative: 5

Brand	Tyre warranty (months)	Price per tyre (Rs.)
A	12	1,200
B	24	1,800
C	36	2,100
D	48	2,700

If the company feels that the warranty period is a good estimate of the tyre life and that a nominal interest rate (compounded annually) of 12% is appropriate, which tyre should it buy? Use Annual Equivalent Method for taking decision.

5.a What do you mean by Commercial Bank? How they are different from Central Bank. Discuss mechanism of credit creation by Commercial Banks. 5

5.b A firm, Prime Manufacturing is planning to expand its production operation. It has identified three machines which are technologically capable to serve the purpose. The initial outlay and annual revenues with each of the machines are given below: 5

	Initial Outlay (Rs.)	Annual revenue	Life (in years)
Machine I	Rs. 5,45,000	Rs. 2,50,000	15
Machine II	Rs. 6,14,000	Rs. 3,30,000	12
Machine III	Rs. 6,00,900	Rs. 3,50,000	10

If the rate of interest is 12%, which machine the company should opt for? Find out the result by Present Worth method.

6.a Discuss similarity and differences between Oligopoly and Monopolistic Competition. 5

6.b A firm's total cost function = $\frac{1}{3}x^3 - 7x^2 + 111x + 50$; and $x = 100 - p$ where p is the price and x is the quantity 5

(i) Write the total revenue function TR
(ii) Find profit maximising level of output
(iii) What is the maximum profit?

7.a Discuss factors which should be considered while deciding location of the firm aboard. 5

7.b The Total Cost (TC) associated with producing and marketing x units of an item is given by $TC = .005x^3 - .02x^2 - 30x + 3000$ 5

(i) Total cost when output is 4 units,
(ii) Average cost of output of 10 units
(iii) Marginal Cost when output is 3 units
(iv) The output where AC =MC

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